EXPERIMENT 11 B : SIMULATION OF RIP USING CISCO PACKET TRACER

AIM:

To configure and simulate Routing Information Protocol (RIP) on routers in Cisco Packet Tracer and verify communication between end devices in different networks.

REQUIREMENTS:

Cisco Packet Tracer, three routers, three switches, three PCs and Ethernet cables.

PROCEDURE:

1. Open Cisco Packet Tracer and create a new project.

2. Place three routers, three switches and three PCs in the workspace.

3. Connect each PC to a switch, and connect each switch to the LAN interface of its router using Ethernet cables.

4. Connect the routers in a line using available router-to-router links. Use serial interfaces if supported, or Ethernet interfaces with suitable addressing.

5. Assign a unique /24 LAN subnet to each router's local network and configure the PCs with an IP address, subnet mask and default gateway.

6. Configure the IP addresses on all router interfaces and enable the interfaces using no shutdown.

7. On each router, enable RIP version 2 and advertise the directly connected LAN and inter-router networks.

8. Use show ip route and show ip protocols to verify RIP configuration and learned routes.

9. Ping PCs in remote networks to test end-to-end communication.

10. Switch to Simulation mode to observe packets travelling through the routers, then save the project.

SAMPLE IP ADDRESSING PLAN:

Network / Device	Interface	IP Address
LAN 1 – PC0	FastEthernet0	192.168.1.10/24
R1 LAN	G0/0	192.168.1.1/24
R1–R2 link	R1 G0/1 / R2 G0/1	10.0.12.1/30 / 10.0.12.2/30
LAN 2 – PC1	FastEthernet0	192.168.2.10/24
R2 LAN	G0/0	192.168.2.1/24
R2–R3 link	R2 G0/2 / R3 G0/1	10.0.23.1/30 / 10.0.23.2/30
LAN 3 – PC2	FastEthernet0	192.168.3.10/24
R3 LAN	G0/0	192.168.3.1/24

Default gateways: PC0 = 192.168.1.1, PC1 = 192.168.2.1, PC2 = 192.168.3.1. Interface names vary by router model; adjust the configuration to match the chosen device.

OUTPUT:

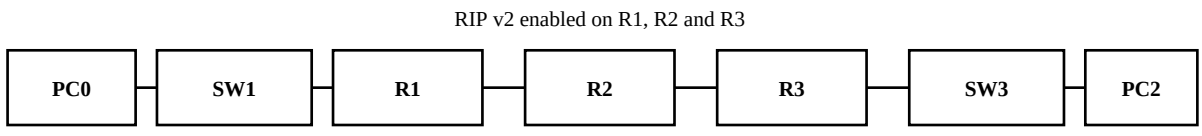

Fig. 1. Example three-router topology. Add PC1 and its switch to the LAN interface of R2.

ROUTER CONFIGURATION:

Router R1:

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.0.12.1 255.255.255.252
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.1.0
Router(config-router)# network 10.0.0.0
Router(config-router)# end
Router# show ip route
Router# show ip protocols
```

Router R2:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.2.0
Router(config-router)# network 10.0.0.0
```

Router R3:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# no auto-summary
Router(config-router)# network 192.168.3.0
Router(config-router)# network 10.0.0.0
```

SAMPLE ROUTING TABLE OUTPUT:

```
R1# show ip route
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
C    10.0.12.0/30 is directly connected, GigabitEthernet0/1
R    192.168.2.0/24 [120/1] via 10.0.12.2
R    192.168.3.0/24 [120/2] via 10.0.12.2
```

CONNECTIVITY TEST:

```
PC0> ping 192.168.3.10

Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126
Reply from 192.168.3.10: bytes=32 time<1ms TTL=126

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

OBSERVATION:

1. RIP version 2 exchanges routing information between routers.

2. Remote networks appear in the routing table with the code R after routes are learned.

3. RIP uses hop count as its metric; the route with the lower hop count is preferred.

4. Successful ping replies confirm that routing between the source and destination LANs is working.

5. Simulation mode can be used to observe packet movement across the routers.

RESULT:

Thus, RIP version 2 was configured and simulated using Cisco Packet Tracer. The routers exchanged network routes, and connectivity between PCs on different LANs was tested using ping.